

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

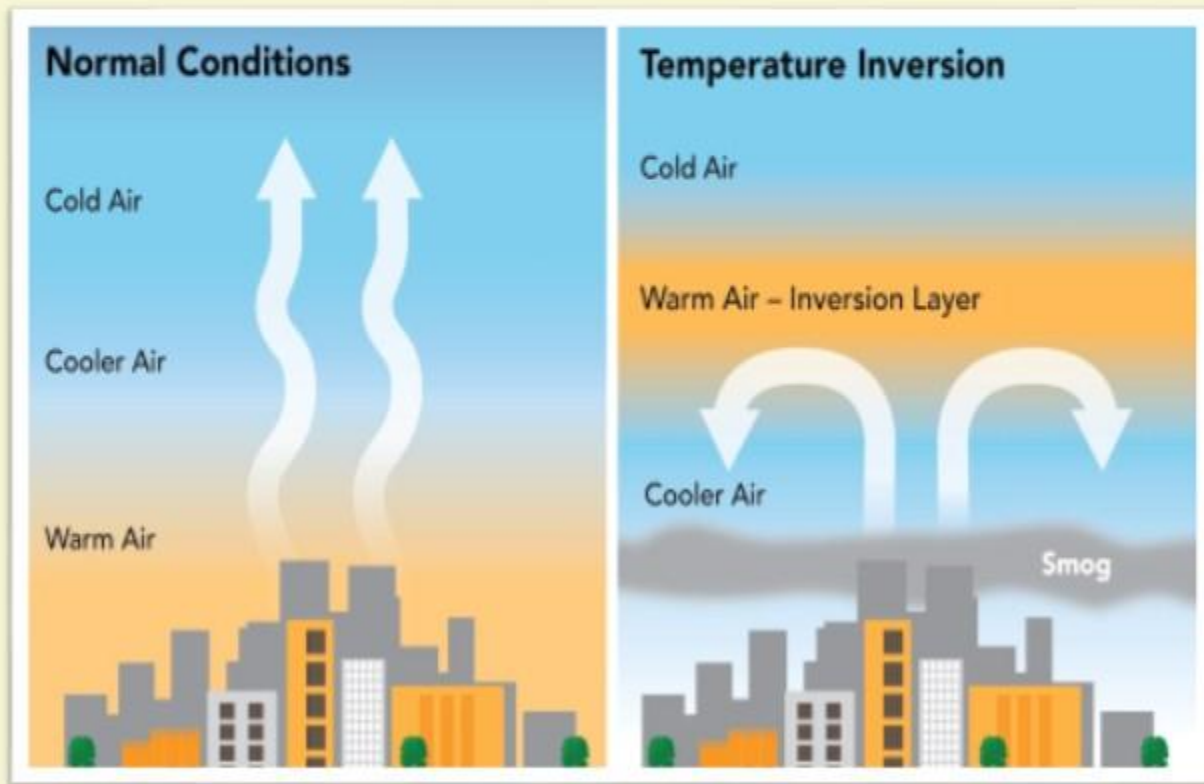
MODULE – 15

TEMPERATURE INVERSION



TEMPERATURE INVERSION

NORMAL CONDITION VS TEMPERATURE INVERSION



- 1 Temperature decreases with increasing height in the troposphere but this varies with seasons, duration of sunshine and location.
- 2 This decrease of temperature takes place at normal lapse of temperature (6.5°C).
- 3 However sometimes temperature **INCREASES WITH INCREASING HEIGHT!!**
- 4 In other words, some times warm air lies above the cold air. This phenomenon is called as **INVERSION OF TEMPERATURE.**



TEMPERATURE INVERSION

NORMAL CONDITION VS TEMPERATURE INVERSION



NORMAL SITUATION

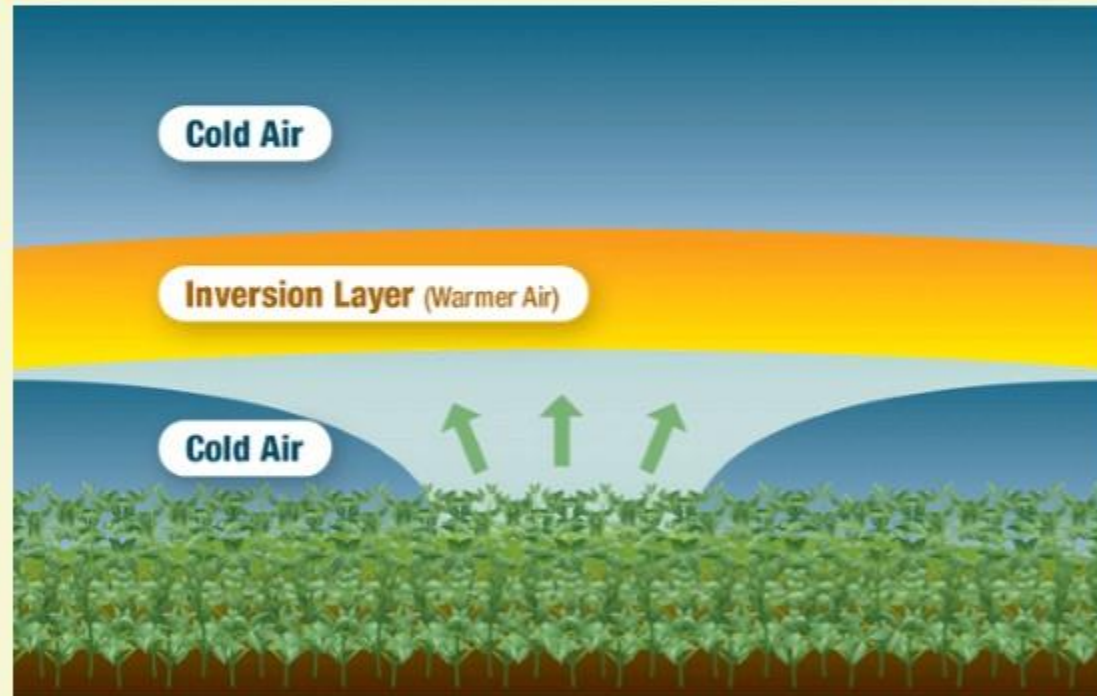
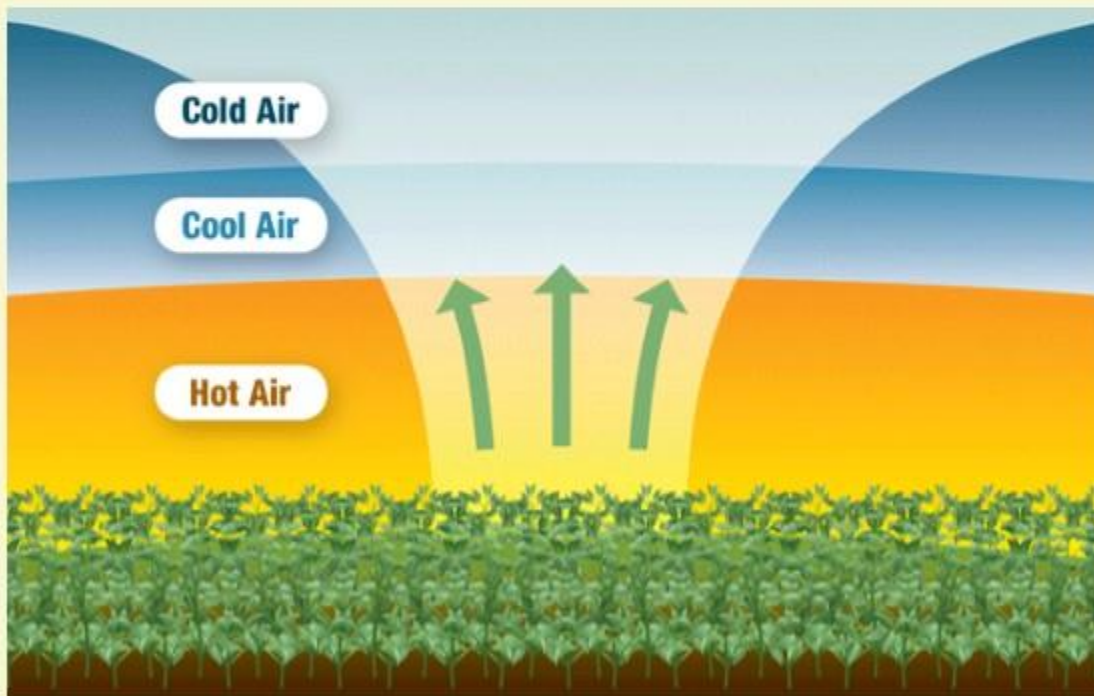


TEMPERATURE INVERSION



TEMPERATURE INVERSION

NORMAL CONDITION VS TEMPERATURE INVERSION





TEMPERATURE INVERSION

NORMAL CONDITION VS TEMPERATURE INVERSION



**UPSC GS-1 MAINS QUESTION
(2013)**



What do you understand by phenomenon of “temperature inversion” in meteorology? How does it affect weather and habitats of the place? (5 Marks)

This question has been asked in previous exams of Geography optional- 1995, 2010.



TEMPERATURE INVERSION

FAVOURABLE CONDITIONS FOR TEMPERATURE INVERSION



1 Long winter night is needed so that the loss of heat by terrestrial radiation from the ground surface during night may exceed the amount of insolation received and thus the ground surface becomes too cold.

- 2**
- Cloudless and clear sky so that the loss of heat through terrestrial radiation proceeds more rapidly without any obstruction.
 - Clouds absorb terrestrial radiation and hence retard loss of heat from the earth's surface.

3 Presence of dry air near the ground surface, so that it may not absorb much heat radiated from the Earth's surface as moist air is capable of absorbing much of the radiant heat from the earth's surface.

4 Static air or slow movement of air, so that there is no transfer and mixing of heat in the lower layers of the atmosphere

5 Snow-covered ground surface, so that there is maximum reflection of insolation.



TEMPERATURE INVERSION

FAVOURABLE CONDITIONS FOR TEMPERATURE INVERSION



QUICK SUMMARY

1. Long winter nights
2. Cloudless and clear sky
3. Presence of dry air
4. Still or very less movement of air
5. Snow-covered ground surface

DO YOU KNOW?

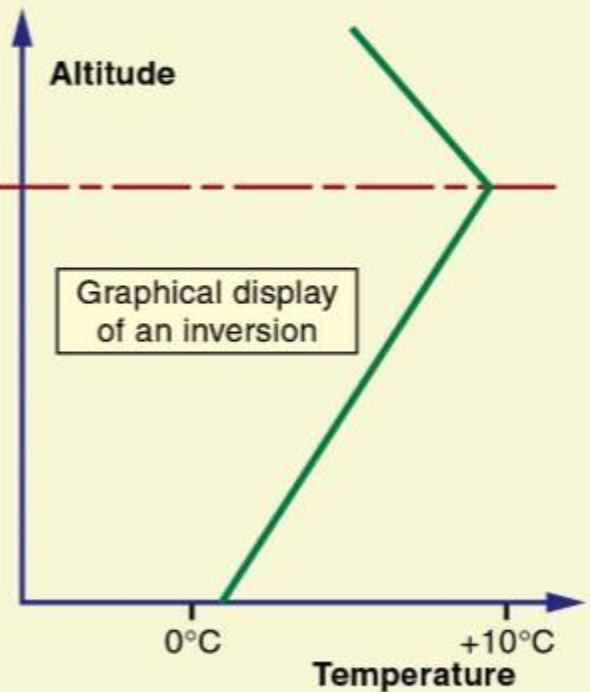
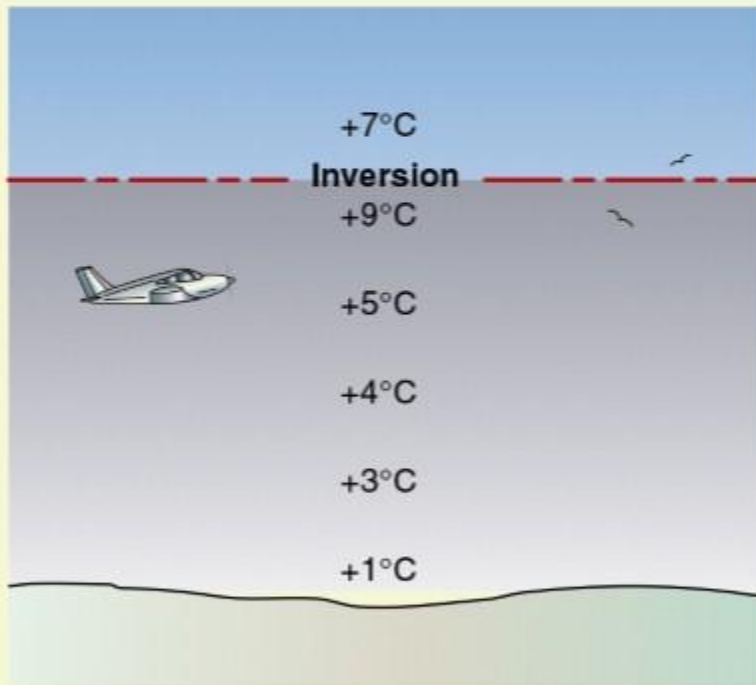
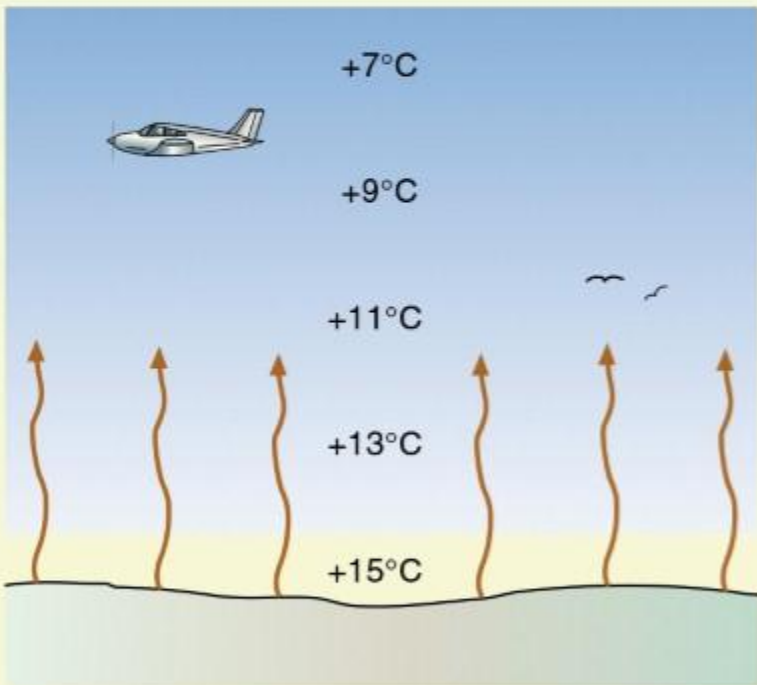


*Ground inversion occurs in the low-latitude areas during winter nights only and the inversion generally disappears with sunrise. It is apparent that **duration and height of inversion increases poleward.***



TEMPERATURE INVERSION

TEMPERATURE VS ALTITUDE GRAPH FOR INVERSION





TEMPERATURE INVERSION

EFFECTS OF TEMPERATURE INVERSION



EFFECTS ON WEATHER

Fog

- Formed due to the situation when warm air lies above cold air
- Low visibility, health hazard
- Affects road/rail/ship transportation
- Favorable for coffee plants in Yemen hills of Arabia where fog protects coffee from direct strong sun rays.

Frost

- If warm air above is condensed at a temperature below freezing point $\Rightarrow$ FROST is formed.
- Definitely economically unfavorable mainly for crops especially fruit orchards.

Dry Conditions

- Fosters stability $\Rightarrow$ restricts upward movement of air



TEMPERATURE INVERSION

EFFECTS ON HABITAT



- Low visibility, health problems, crop failure due to frost formation.
- People settle in upper slopes in hilly areas & not in valleys
- Farmers have to use special fans & heaters to protect their orchards from frost bites. Eg: Californian & Switzerland orchards
- Forms SMOG. Air pollution / smoke + Fog = SMOG

Previously Seen In Module 12,

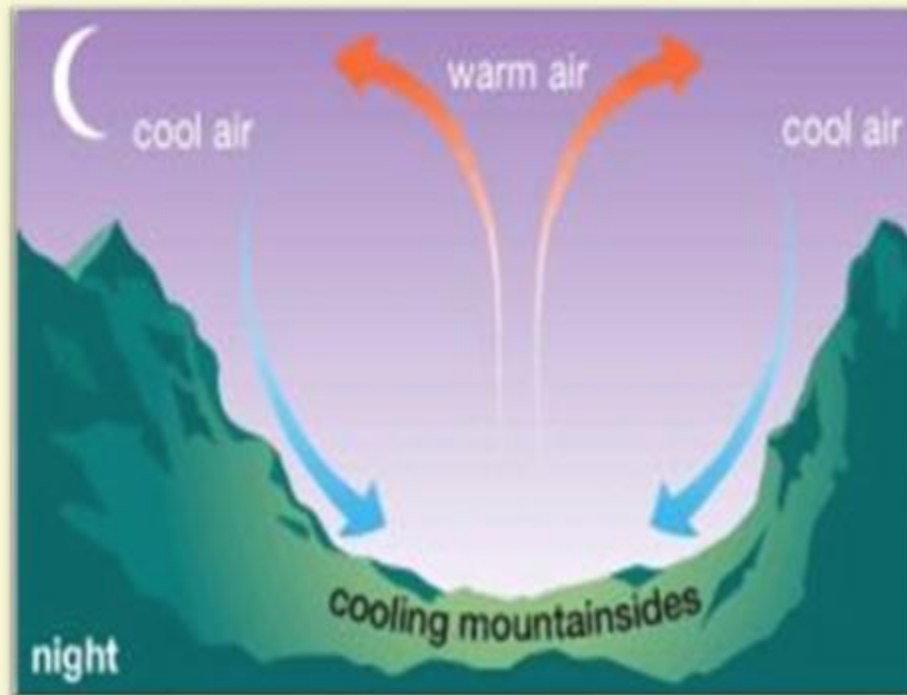


FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 6

PREVAILING WINDS



MOUNTAIN BREEZE

- 1** In the night, the **mountain top** cools up faster creating a **thermal high**.
- 2** This creates a **downslope movement** of **cool air** from the hill top towards the valley known as the **mountain breeze**.



TEMPERATURE INVERSION

EFFECTS OF TEMPERATURE INVERSION



How is smog formed?

In the winter, cooler temperatures and dry winds create a high-pressure cool layer in the atmosphere

Warm air aloft acts as a lid that traps smog closer to the ground

Cool air

Warm air

Cool air

TRAPPED SMOG
Pollutants react with sunlight to create photochemical toxins



Pollution from vehicles, factories, fires and other human activities releases air pollutants

WHAT IS SMOG?

- 1 Smog is a **yellowish or blackish fog** formed mainly by a mixture of pollutants in the atmosphere which consists of **fine particles and ground level ozone**.

HOW IS SMOG FORMED?

- 1 During the **winter months** when the **wind speeds are low**, it helps the **smoke and fog** to stagnate at a place forming smog.
- 2 The time that smog takes to form depends directly on the temperature. **Temperature inversions** are situations when warm air does not rise and instead stays near the ground. This is **the ideal situation for smog formation**.



TEMPERATURE INVERSION

EFFECTS OF TEMPERATURE INVERSION

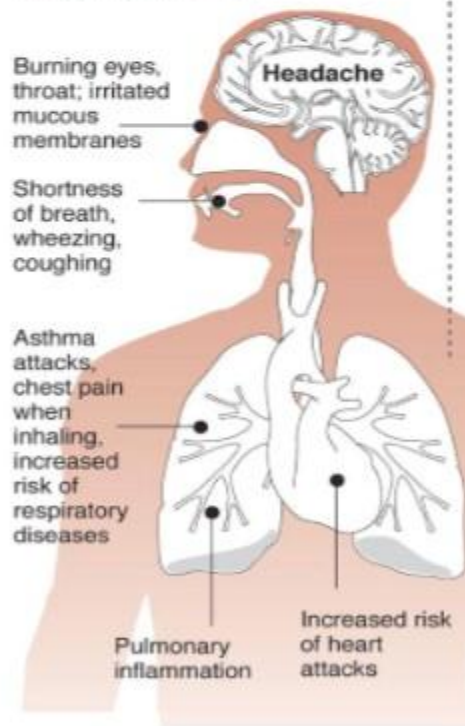


HARMFUL IMPACTS OF SMOG

Why smog is harmful

Ozone, the main ingredient in smog, is one of the most widespread air pollutants and among the most dangerous.

Effects on health



How ozone forms

1 Oxygen in the atmosphere O_2



2 Nitric oxide, byproduct of combustion



3 Sunlight breaks up nitric oxide



4 Ozone formed by three oxygen atoms



U.S. ozone limits

In parts per billion

• 1997-2008 **84**

• 2008-present **75**

• New EPA proposal **60-70**

1

Causes health problems like

- asthma
- chronic bronchitis
- eye irritation
- reduced resistance to colds and lung infections

2

Inhibits plant growth and can cause widespread damage to crops and forests



CASE STUDIES (TO BE USED IN MAINS ANSWERS)



The Great Smog of London in 1952 was caused by using coal as the main source of heat and was responsible for the deaths of around 12,000 individuals.





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NORMAL CONDITION VS TEMPERATURE INVERSION



UPSC MAINS 2013 QUESTION



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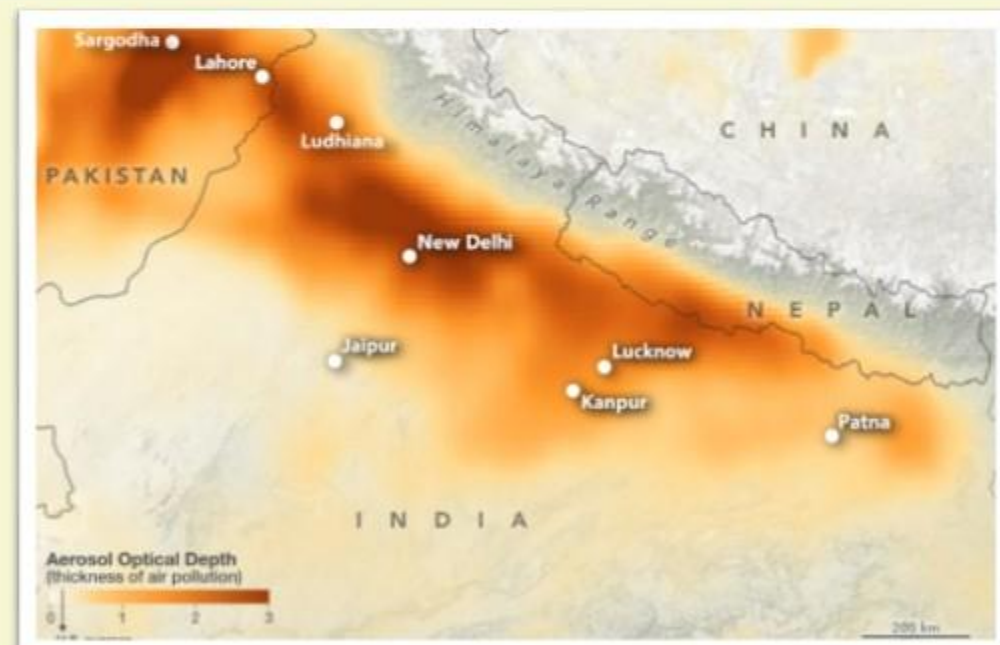
TEMPERATURE INVERSION

EFFECTS OF TEMPERATURE INVERSION



PROBABLE QUESTION FOR MAINS

What is temperature inversion? Examine some reasons for the persistent smog in Delhi in winters in recent times.



Smog in North India

WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

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